AfyaSasa Pharmacy — backend audit, stock-take verification and completion report

Date: 22 October 2026

Tenant: `jalaram` · **schema:** `demo` · **API:** `/api/v1`

HEAD: `247d026` on `main`

No commit, push, merge or deploy.

1. Executive summary and release decision

CSV stock-take posting no longer bypasses independent approval. Counts can be imported into an open session; posting still requires a different authorised user. Physical count sheets use `book_qty` and a blank `physical_count` column.

The local compose database is **not disposable** (26 patient records, 66 inventory ledger rows). Mutating stock-take posting, sales, dispensing and recount were **not** run against it.

Authorised login on `http://localhost:8080/` **succeeded**. Pharmacy, POS and stock-take screens were opened and photographed. Full count → approve → post was not executed.

Controlled release: no. Staff-in-the-loop local UI: yes, with the caveats below.

2. Repository baseline (preserved)

Uncommitted pharmacy/POS/HCC/error-handling work was left intact. Additional changes this pass: CSV post disabled, session sheet + import-counts APIs, UI copy, tests, this evidence pack.

Compose backend: `POSTGRES_HOST=postgres`, `POSTGRES_DB=afyasasa`, `TYPEORM_MIGRATIONS_RUN=false`, `AFYASASA_ALLOW_DEMO_SEED=false`.

Postgres container `/afya-sasa-postgres-1` created 2026-09-21. Host `.env` `POSTGRES_PORT=5432` is **not** the compose map (`5433`).

3. Architecture and data flow

One ledger: `demo.inventory_transactions`. Holds: `demo.inventory_reservations` (`held` / `consumed` / `released`). Charges/payments unchanged. Walk-in: nullable `patient_id` on pharmacy charges only.

Prepare → hold → cashier → `evaluatePharmacyClearance` → issue `DISPENSE`. Cancel releases holds; it does not refund.

4. Schema and migrations

Object	Role
`demo.inventory_items` / `batches` / `locations`	Catalogue and on-hand
`demo.inventory_transactions`	Ledger
`demo.inventory_reservations`	Holds
`demo.inventory_stock_take_sessions` / `_lines`	Count lifecycle

`demo.charges` / `payment_transactions`	Cashier

Counts at inspection: items 2, batches 13, ledger 66, charges 0, patients 26. **Not a throwaway database.** No new migrations applied this pass.

5. Endpoint / service / table map (stock-changing)

Route	Service	Writes
`POST /inventory/receive`	`receiveStock`	RECEIPT + on-hand
`POST /inventory/items/import` confirm receipt	`importCatalog`	RECEIPT import
`POST /inventory/items/import` confirm stock_take	`importCatalog`	**Rejected** — `STOCK_TAKE_INVALID`
`POST /inventory/stock-take-sessions`	`createStockTakeSession`	Session + baseline lines only
`POST .../lines/:id` and `.../import-counts`	count / import	Line counts only
`POST .../approve`	`approveAndPostStockTakeSession`	`STOCK_TAKE` ledger, once
`POST /inventory/sales/prepare`	prepare Rx/OTC	Reservations + charge
`POST /inventory/dispense/*`	issue	Consume hold + DISPENSE
`POST /inventory/sales/cancel`	cancel	Release hold
Transfers ship/receive	transfer methods	TRANSFER_OUT/IN

6. Stock-take and physical sheet

Session: location, `baseline_at`, lot lines, `expected_qty`. Counting does not change on-hand. Variance = counted − book. Reason required if variance ≠ 0. Submit all lines. Reject keeps baseline. Approver ≠ submitter. Stale on-hand aborts post. Session id is the ledger reference.

Sheet: `GET /inventory/stock-take-sessions/:id/sheet`

Columns: `session_id, line_id, sku, name, batch_no, expiry, book_qty, physical_count, variance_reason`

`physical_count` is empty. `book_qty` is not aliased as counted.

Import: `POST .../import-counts` writes counts into the session only.

7. Variance and approval

Unit tests: reason required on difference; count blocked after submit/post; independent approver; CSV confirm cannot post.

8. Legacy CSV bypass — finding and fix

Finding: `importCatalog` posted `STOCK_TAKE` when `confirmStockTake` was true, with only `inventory:manage` and a client `sessionId`. That bypassed independent approval.

Fix: stock_take + confirm now throws before any transaction. Regression test asserts the manager transaction is not opened. UI Confirm-and-post removed.

Receipt imports are unchanged (goods receipt, not stock-take).

9. POS, payment, dispensing

POS search on live app returned Paracetamol PARA500, 4694 available, **no price**. Prepare/issue buttons stay disabled on an empty cart. Cashier ledger tile: collected 0 / charged 0. No mutating sale was posted.

Walk-in fields are on the POS screen. Clinical patient search remains separate.

10. Permissions

Count/submit/import-counts: `inventory:manage`. Approve: `inventory:manage` or `settings:manage`, different user. Oversight: `inventory:read`. Dispense: `pharmacy:dispense`. Stock Oversight does not require `settings:manage`.

11. Files changed this pass (plus preserved dirty tree)

- `inventory.service.ts` / `inventory.controller.ts` / `inventory.dto.ts` — reject CSV stock-take post; sheet + import-counts
- `inventory.service.spec.ts` / `stock-take-session.spec.ts`
- `StockTakeWorkspace.tsx` / `StockTakeSessionPanel.tsx`

12. Migrations applied this pass

None.

13. Tests

```
npm --prefix backend test -- --runInBand --forceExit
```

87 suites passed, 470 passed, 20 skipped, 0 failed.

```
npm --prefix backend run build
npm --prefix frontend run build
```

Both succeeded. Images rebuilt. `inventory-pg` **NOT RUN** (no proven disposable `inventory_it`).

14. Browser scenarios

Tool: Cursor browser at `http://localhost:8080/` after image rebuild. No screen recording in this tool.

#	Scenario	Result	Evidence
1	Login (authorised staff)	PASSED	01, then pharmacy home
2	Pharmacy navigation	PASSED	02
3	POS screen	PASSED	03
4	Catalogue / available qty	PASSED (a11y) / PARTIAL (screenshot below fold)	04 + tree PARA500 4694
5	Stock-take session UI	PASSED	05
6	Create session / sheet / count	NOT RUN	Would lock a live store
7	Variance / submit / reject / independent approve / post	NOT RUN	Same

8	CSV import through session	NOT RUN on DB	Unit: CSV post blocked
9	Prepare / pay / issue / unpaid block / walk-in sale	NOT RUN	Non-disposable DB
10	Stock Oversight overview	BLOCKED	Nav click intercepted
11	Recording	NOT RUN	Tool has no recorder

15. Limitations

- Operational local DB — no synthetic mutating E2E.
- POS items with no sell price.
- Sheet-import header on the running image still mentioned “direct posting” until the copy fix (backend already rejects it).
- No second inventory user in-browser for independent approval.
- Recall/quarantine/disposal, insurance, POS tax/discount still fail-closed / not implemented.

16. Backup / recovery

Do not `docker compose down -v`. Existing `ops/backup-postgres.sh`. Stock-take session tables already exist; dropping them is not required for this fix.

17. Checklist

Item	**Status**
Uncommitted work preserved	Yes
CSV stock-take bypass removed	Yes (unit verified)
Physical sheet columns	Implemented; download not exercised (no session)
Independent approval in backend	Yes (unit)
Browser login	PASSED
Mutating stock-take / sale E2E	BLOCKED
Isolated integration DB	BLOCKED
Controlled release	**No**

Evidence: `docs/pharmacy-e2e-evidence-2026-10-22/`

This file: `docs/pharmacy-backend-and-e2e-audit-2026-10-22.md`